DAY 4 - BEGINNER

IF Logic for Data Classification

Primary tool: Google Sheets

What you are learning

Boolean thinking: condition, result if true, result if false.

Learning objectives

- Create a simple IF

- Translate a written rule into logic

- Test boundary values

Worked practical

Classify learners as Pass/Fail and Eligible/Not Eligible using stated thresholds.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

Give AI the business rule and your formula; ask it to critique the logic and test edge cases.

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

Create rule-based status fields.

Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.